

CS 4720 HW9

1a) $P(\text{toothache}) = 0.108 + 0.012 + 0.072 + 0.008 = 0.20$

b) $P(\text{cavity}) = 0.108 + 0.012 + 0.072 + 0.008 = 0.20$

c) $P(\text{toothache}|\text{cavity}) = P(\text{Toothache} \wedge \text{cavity}) / P(\text{cavity}) = (0.108 + 0.012)/0.2 = 0.120/0.20 = 0.6$

d) $P(\text{Cavity} | \text{toothache} \vee \text{catch}) = P(\text{Cavity} \wedge (\text{toothache} \vee \text{catch})) / P(\text{toothache} \vee \text{catch})$

Let us first calculate the numerator over here, where we know that the only section that we are going to exclude in this case is going to be the last one ($\neg\text{toothache} \wedge \neg\text{catch}$) since we need at least one of them to be true, so we can say that the numerator is going to be $0.20 - 0.008 = 0.192$.

Now in order to calculate the denominator, we can use the rule $P(A \vee B) = 1 - P(\neg A \wedge \neg B)$ so that means we need to calculate the $P(\neg\text{toothache} \wedge \neg\text{catch})$ first, which is $0.576 + 0.008 = 0.584$ (the last column) so the $P(\text{toothache} \vee \text{catch}) = 1 - 0.584 = 0.416$, so now we can do the following calculation to take out the answer:

$$P(\text{Cavity} | \text{toothache} \vee \text{catch}) = 0.192/0.416 = 0.46 \text{ (approx)}$$

2) Let us use the Bayes Rule to solve this question.

Quantity	Meaning	Test A	Test B
$P(v)$	It means that a person actually has the virus	0.01	0.01
$P(\neg v)$	It means that a person does not have the virus	0.99	0.99
$P(a v)$	It represents the true positive rate of test A	0.95	NA
$P(b v)$	It represents the true positive rate of test B	NA	0.9
$P(a \neg v)$	It represents the false positive rate of test A	0.10	NA
$P(b \neg v)$	It represents the false positive rate of test B	NA	0.05

$$P(v|a) = P(a|v) \cdot P(v) / (P(a|v) \cdot P(v) + P(a|\neg v) \cdot P(\neg v)) = (0.95 \cdot 0.01) / ((0.95 \cdot 0.01) + (0.10 \cdot 0.99)) = 0.8776 \text{ (approx)} = 8.8\%$$

$$P(x|b) = P(b|v) \cdot P(v) / (P(b|v) \cdot P(v) + P(b|\neg v) \cdot P(\neg v)) = (0.90 \cdot 0.01) / ((0.90 \cdot 0.01) + (0.05 \cdot 0.99)) = 0.154 \text{ (approx)} = 15.4\%$$

Since, $15.4\% > 8.8\%$, we can say that test B is more indicative about telling whether a person truly has the virus or not.

3) Since the disease is rare, we can say that most of the test results could be false positive, so we do not need at panic and that's why it is a good news that the disease is rare.

In the second part of the question we need to compute the chances that we actually have the disease that is we need to calculate $P(\text{Disease} | \text{Positive})$

$$P(\text{Disease} | \text{Positive}) = P(\text{Positive} | \text{Disease}) \cdot P(\text{Disease}) / (P(\text{Positive} | \text{Disease}) \cdot P(\text{Disease}) + P(\text{Positive} | \neg \text{Disease}) \cdot P(\neg \text{Disease}))$$

According to the question, we have the following values:

$$P(\text{Disease}) = 1/10000 = 0.0001$$

$$P(\neg \text{Disease}) = 1 - 0.0001 = 0.9999$$

$$P(\text{Positive} | \text{Disease}) = 0.99$$

$$P(\text{Positive} | \neg \text{Disease}) = 0.01 \text{ (false positive rate)}$$

Now, let us substitute these values in the equation:

$$P(\text{Disease} | \text{Positive}) = 0.99 \cdot (0.0001) / ((0.99 \cdot 0.0001) + (0.01) \cdot (0.9999)) = 0.01 \text{ (approx)}.$$

Hence we can say that even if we test positive, there is approximately 1% chance that we actually might have the disease.